

AI PROMPTVERSE

National Level AI Hackathon

PROBLEM STATEMENT

AI for Sustainability

Challenge: “Use Intelligence to Build a More Sustainable Future.”

Background

Environmental challenges such as increasing waste, rising carbon emissions, excessive consumption, and food wastage require smarter and more accessible solutions. Artificial Intelligence and Large Language Models (LLMs) can help individuals, businesses, and communities make better, more sustainable decisions.

Your Challenge

Design and develop an **AI-powered solution** that addresses a real-world sustainability problem and encourages environmentally responsible behaviour.

Possible Areas to Explore

- **Smart Waste Classification** – Use AI/computer vision to identify and classify waste and suggest the appropriate disposal or recycling method.
- **Personal Carbon-Footprint Assistant** – Estimate an individual's carbon footprint based on lifestyle choices and provide actionable ways to reduce it.
- **Smart Recycling Assistant** – Help users determine whether an item can be reused, recycled, composted, or safely disposed of.
- **Sustainable Product Recommendations** – Recommend environmentally friendly alternatives based on product materials, lifecycle, packaging, and usage.
- **Food-Waste Reduction** – Predict food requirements, track expiry dates, suggest recipes using available ingredients, or help reduce unnecessary food wastage.

Expected Outcome

Build a **functional AI/LLM-powered prototype** that demonstrates how technology can contribute to environmental sustainability. The solution should ideally provide meaningful recommendations, predictions, automation, or decision-making assistance rather than simply presenting information.

Key Evaluation Focus

AI Innovation • Sustainability Impact • Practicality • User Experience

“Use intelligence to build a more sustainable future.”